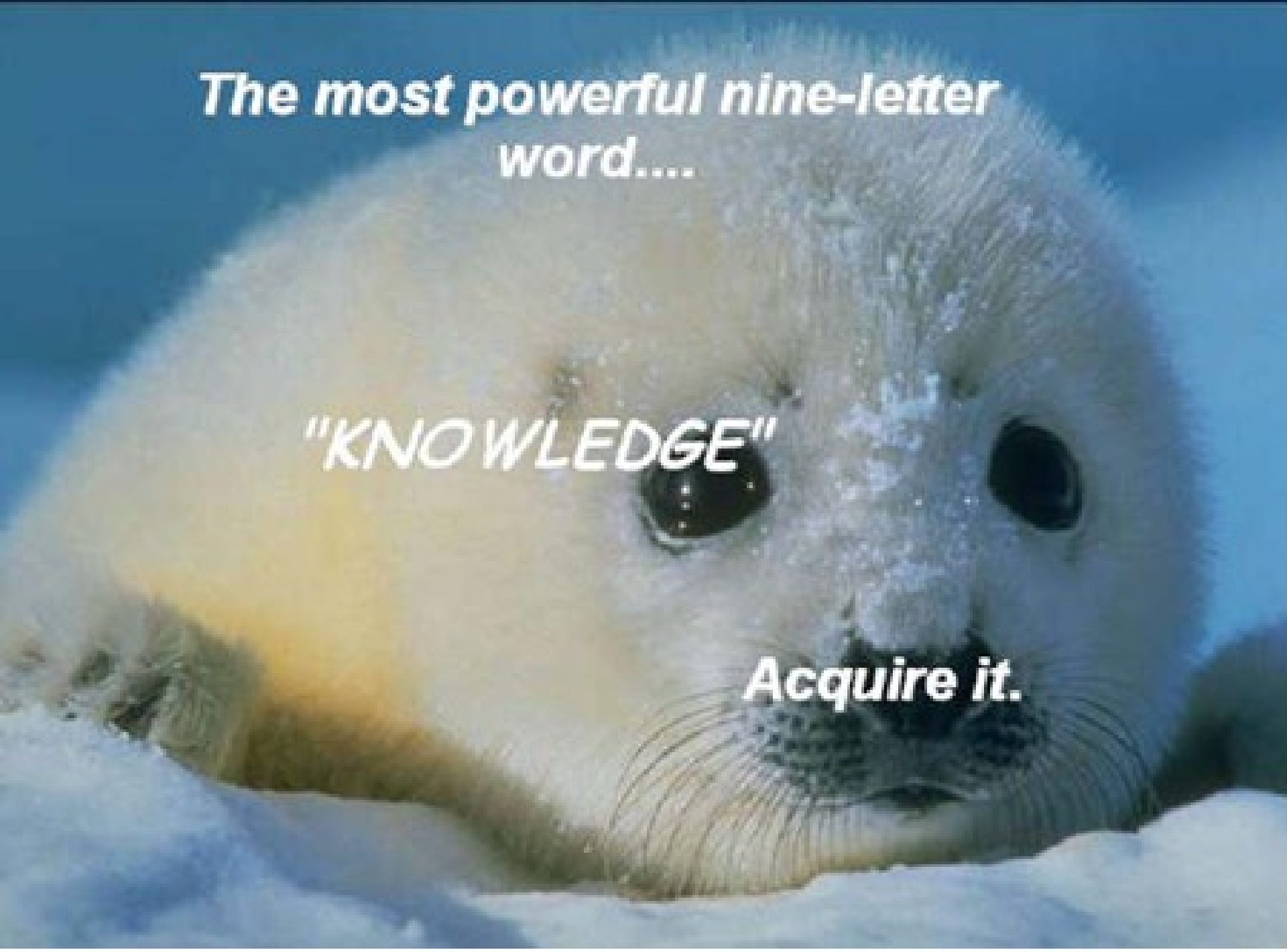

The hardest working seven-letter word.
"SUCCESS"
Achieve it.


The most powerful nine-letter word....
"KNOWLEDGE"
Acquire it.

LIPID DISORDERS

Hyperlipidemia is associated with increase risk of atherosclerosis related diseases like IHD and stroke

LIPID AND LIPOPRTEIN METABOLISM

- Major plasma lipids—cholesterol
  TG
  phospholipids
  FFA
- Cholesterol is a major component of cell membranes and a precursor of steroid hormones and bile acids

- TG are bodies major energy store particularly in adipose tissues
- Cholesterol + TG are relatively insoluble in plasma and are transported as lipoproteins
- Nomenclature of the lipoprteins is based on their separation by density gradient

- Because lipid molecules (cholesterol and TGs) are water insoluble, they must be packaged in special molecular complexes known as lipoproteins in order to be transported in plasma. Lipoproteins may accumulate in the plasma due to overproduction and/or deficient removal

- HDL contains the smallest densest particles and lowest content of lipid
- Chylomicrons have a high content of lipid , are large and least dense
- Lipoproteins transport absorbed dietary fat and endogenously synthesised cholesterol and TG

Lipoprotein pathway is simplified in 3 main pathway

1--Exogenous pathway---lipids from food

2--Endogenous pathway ---lipids synthesised by the liver

3--Reverse cholesterol transport---return of cholesterol from tissues to liver

- In the cells of the small of intestine, dietary lipids are packaged into large lipoprotein complexes called **chylomicrons**, which have a high-TG, low-cholesterol content.
- Chylomicrons reach the circulation via lymph, and are subject to the action of lipoprotein lipases in the blood capillaries of muscle and adipose tissue, where TGs are hydrolyzed and taken up. The remaining cholesterol-rich structure, now call **chylomicron remnant** is taken up by the liver cells and broken down into its individual components.
- Some pts may have a lipoprotein lipase deficiency resulting in elevated chylomicron concentration (**type I** hyperlipidemia). There is no evidence that chylomicrons are pro-atherogenic; they are probably too large to penetrate the vascular endothelium.

Lipoproteins are

1- chylomicrons

2-VLDL—very low density lipoprotein

3-LDL—low density lipoprotein

4- HDL –high density liporotein

- Cholesterol

  -- A steroid alcohol synthesized in the liver with variable amounts obtained from diet

  -- Average total body cholesterol = 150 g of which 90% as part of cell membrane structure.

  -- Necessary for the synthesis of steroid hormones

- **Triglycerides** (TGs; fat) are used by muscle cells for energy production or taken up and stored in adipose tissue. Excess food intake is converted to fat (excess calories Þ TGs) and deposited in adipose tissue to be mobilized when the need arises (i.e., when energy expenditure exceeds intake).

- **Triglycerides (TGs)**

- -- Saturated fat: red meat, dairy products, coconut oil and palm oil.
  -- Unsaturated fat: :
- (fish oil, soybean, canola oil)
- : vegetable oils (corn, sunflower)
  --Monounsaturated fat: olive oil Trans-fatty acids: hydrogenated vegetable oils (margarine

- **VLDL**
  -- Produced in the liver , Rich in TGs (>65%) and cholesterol (20%)
  -- Serves to transport endogenous lipids (particularly TGs) to extrahepatic sites.
  ------Hydrolyzed by LPL first to IDL and then to LDL

- **IDL**
- Derived from both VLDL and HDL.
  --IDL particles are either taken up by the liver, a process mediated by apoE or may be reduced to LDL (by losing apoE and more TGs).

- **LDL** -- Product of VLDL catabolism (catalyzed by LPL)
  -- Contains >60% cholesterol and accounts for >60% of the total plasma cholesterol.
  -- Taken up by hepatic and extrahepatic tissue through receptor-mediated endocytosis triggered by apoB100 - LDL receptor interaction.
  -- Has a high atherogenic potential. LDL activates monocytes (Þ macrophages). Oxidized LDL is a major source of cholesterol for macrophages in atheromatous plaques.

- **HDL**
- -- Produced by extrahepatic tissue and serves as a vehicle for the transfer of cholesterol from the peripheral tissues to the liver.
  -- HDL takes up the cholesterol liberated in the course of normal cell membrane turnover.
  -- Cholesterol esters are transferred form HDL to IDL which are then converted to LDL or take up by the liver directly.

DYSLIPOPROTEINEMIA

Lipid disorders are classified as—

- Primary
- secondary

A. Primary Hyperlipoproteinaemias:

Frederickson *et al. (1967) proposed five types based*
on changes in plasma lipoproteins.

1. Type-I: Familial Lipoprotein Lipase Deficiency:

- A rare disorder and is characterised by Hypertriglyceridaemia (TG ↑), and hyperchylomicronaemia.

Chylomicrons grossly increased ↑↑ and there is slow clearing of chylomicrons.

- VLDL (Pre-β-Lipoproteins) also increased, more

so in, increased carbohydrate intake.

- Decrease in α-lipoprotein (HDL↓) and β lipoproteins (LDL↓).

Inheritance:* *Autosomal recessive

Enzyme deficiency:

• Deficiency of the enzyme ***lipoprotein lipase.***

• A variant of the disease can be produced by deficiency of apo-C II.

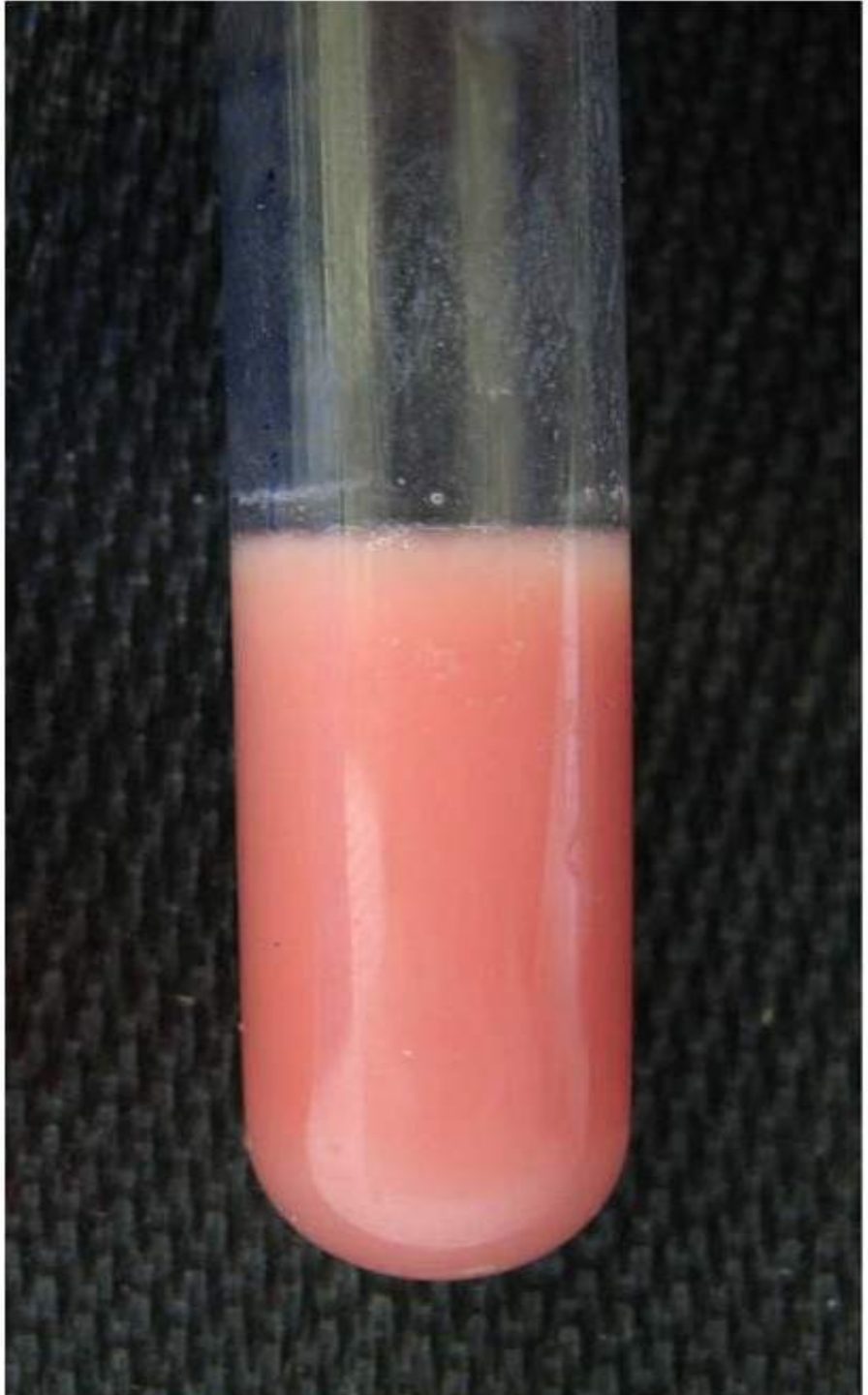
Clinical feature: Presents in early childhood and is characterised by

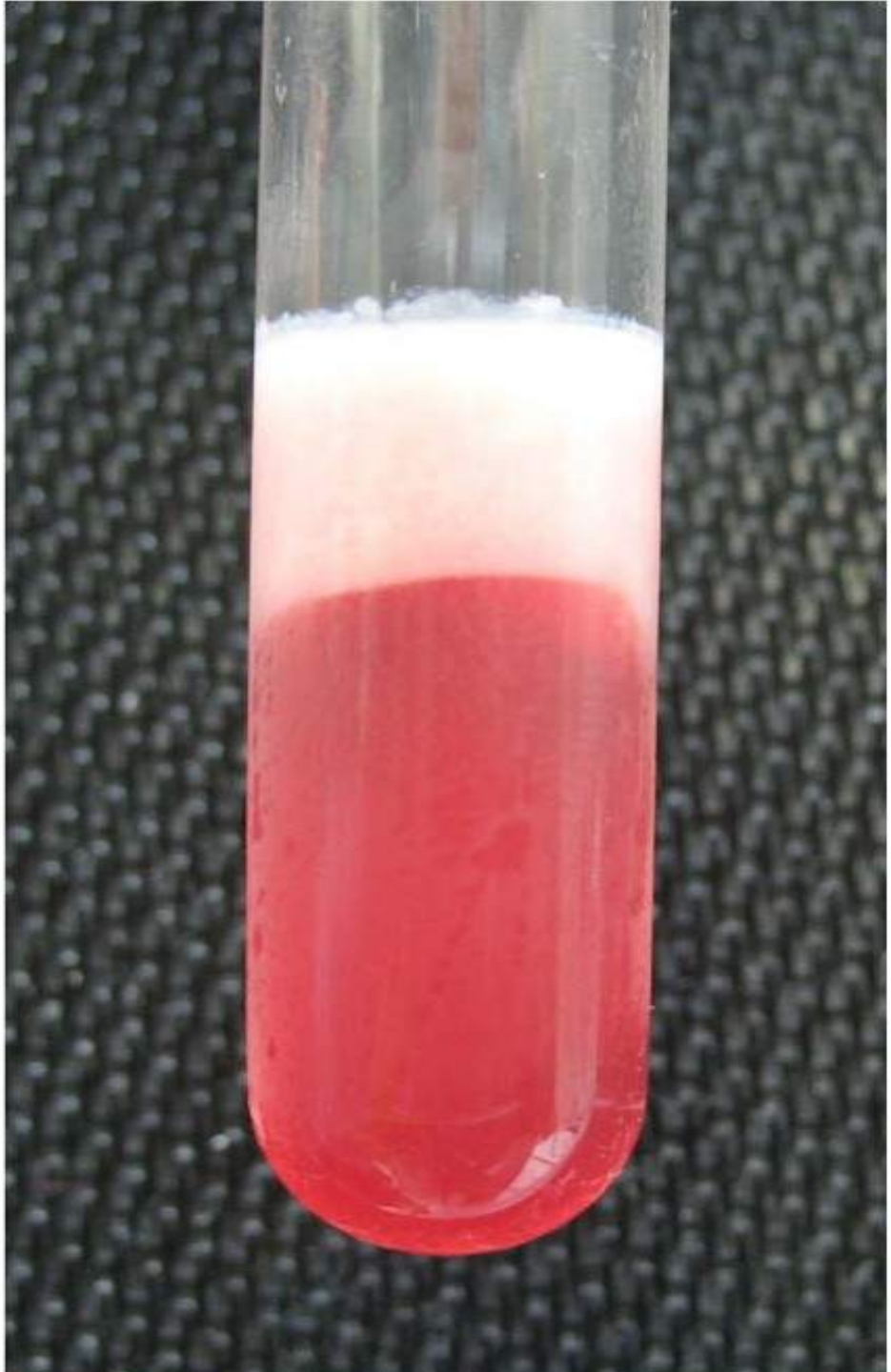
• Eruptive Xanthomas

• Recurrent abdominal pain

Refrigeration test:

If serum of suspected patient is taken in a narrow small tube and kept in refrigerator temperature for 24 hrs undisturbed, a clear zone of chylomicrons is seen to float on the top and make a distinct separate layer





2. Type II: Familial Hypercholesterolaemia (FHC):

A common disorder which has been extensively investigated.
The disease is characterised by:
• Hyper β-lipoproteinaemia (LDL ↑)
• Associated with increased total cholesterol ↑
• VLDL may be raised, hence total TG may be high.
But plasma usually remains clear.

Inheritance: Autosomal dominant;

Frequency—0.2 percent

Enzyme deficiency (metabolic defect): There is no enzyme deficiency. Metabolic defects are:

- An increased synthesis of apo-B ↑
- Defective catabolism of LDL. Deficiency of LDL receptors in fibroblasts demonstrated.

Clinical features: (a) Xanthomas of tendinous and tuberous type have been described,

(b) Corneal arcus,

(c) Occasionally xanthelesma.

Clinically most important is the ***increased incidence of atherosclerosis and premature cardiovascular diseases***



3. Type-III: Familial Dys-Beta Lipoproteinaemia:

Synonyms: Broad Beta disease, 'Remnant' removal disease

The disease is characterised by:

- Increase in β-lipoproteins (LDL ↑)
- Increase in Pre-β-lipoproteins (VLDL ↑);
- Actually rise is in IDL (VLDL remnant). This appears as **broad β-band *("floating" β-band)-β-*** VLDL on electrophoresis. Also there is hypercholesterolaemia and hypertriglyceridaemia.

Inheritance: Autosomal dominant.

:Metabolic defects:

• Increased concentration of apo-E ↑.
• Increased synthesis of apo-B ↑.
• Conversion of normal VLDL to β-VLDL (IDL) and its degradation without conversion to LDL. Defect is in “Remnant” metabolism.

Clinical feature: Presents with Xanthomastuberous and palmar Xanthomas. Premature cardiovascular diseases and atherosclerosis common. Also prone to develop peripheral vascular diseases

Treatment

• By weight reduction
• Low carbohydrate diets, containing unsaturated fats and little cholesterol.


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4. Type-IV: Familial Hypertriglyceridaemia (FHTG):

The disease is characterised by:

- Hyper Pre-β-lipoproteinaemia (VLDL ↑)
- With increase in endogenous synthesis of TG ↑
- Cholesterol level may be normal or increased
- α and β-lipoproteins are subnormal (HDL↓, LDL↓)

Inheritance: Autosomal dominant.

Metabolic Defects:

- Increased endogenous synthesis of TG ↑
- Decreased catabolism of both
- Glucose intolerance is frequent.

Clinical features: Usually presents in early adulthood.

The above-mentioned lipoprotein pattern is also found:

- In association with coronary heart disease (CHD)
- With maturity onset Diabetes mellitus
- In obesity
- With chronic alcoholism, and
- With taking of progestational hormones

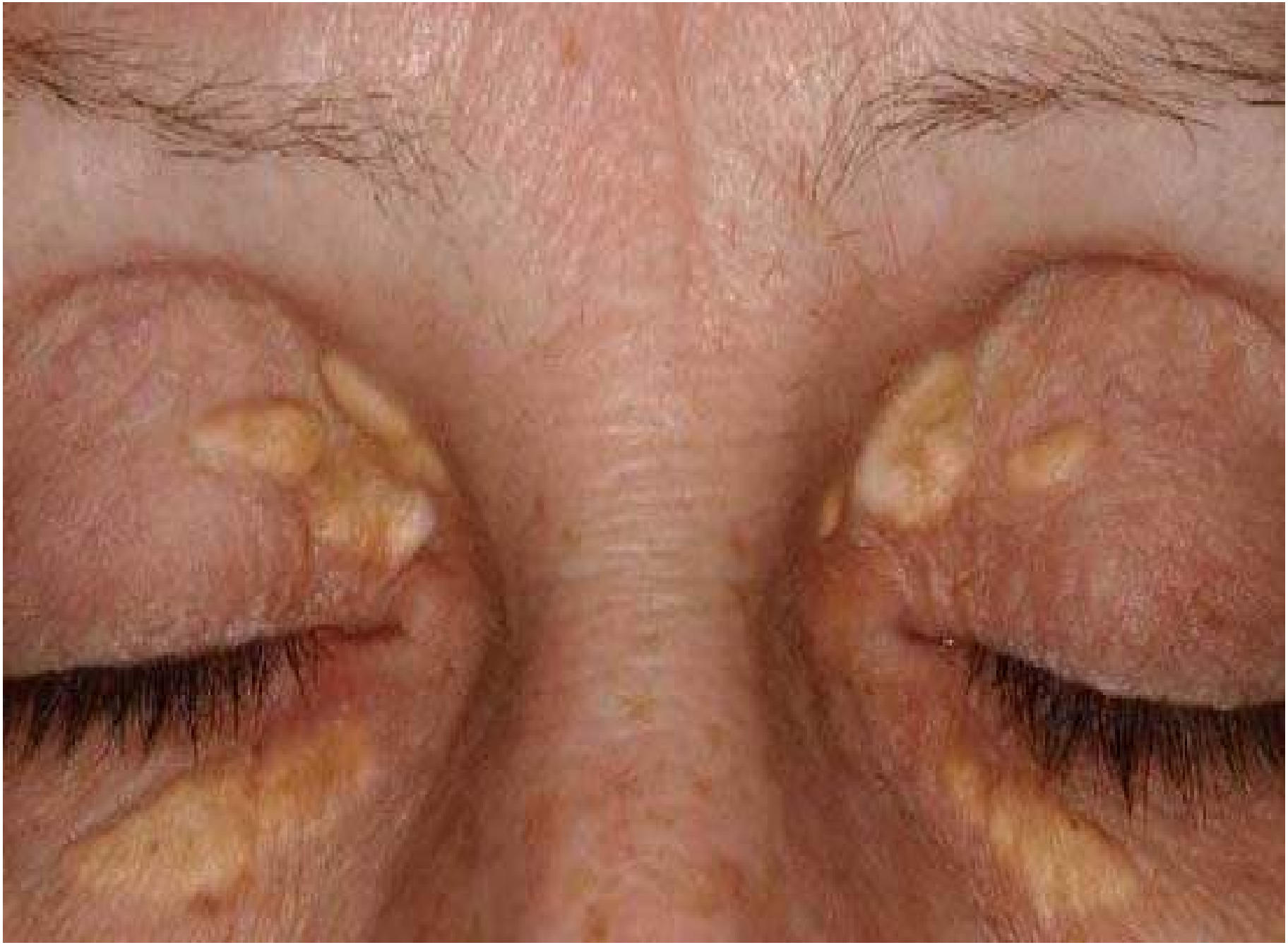
Treatment:

- Weight reduction,
- Decrease of carbohydrate intake in diets with unsaturated fats
- Low cholesterol diet
- Use of hypolipidaemic agents

5. Type-V: Combined Hyperlipidaemias: In this disease, the lipoprotein pattern is complex. Increase in both chylomicrons and Pre-β-lipoproteins (VLDL) are seen. Hypertriglyceridaemia (TG ↑), Hypercholesterolaemia?
Concentration of α-lipoproteins
(HDL) and β-lipoproteins (LDL) are decreased.

Inheritance: Autosomal dominant.

Clinical feature: The disorder is manifested only in adulthood. Xanthomas are frequently present; incidence of atherosclerosis is not striking. Glucose tolerance is usually abnormal. Frequently found associated with obesity and Diabetes mellitus. The reason for the condition being familial is not known.



Treatment:

- Weight reduction
- Followed by diet not too high in carbohydrates and fats.

WOLMAN'S DISEASE: Also called cholesteryl ester storage disease. Increased levels of cholesterol is seen (hypercholesterolaemia).

Enzyme deficiency: Due to a deficiency of cholesteryl ester

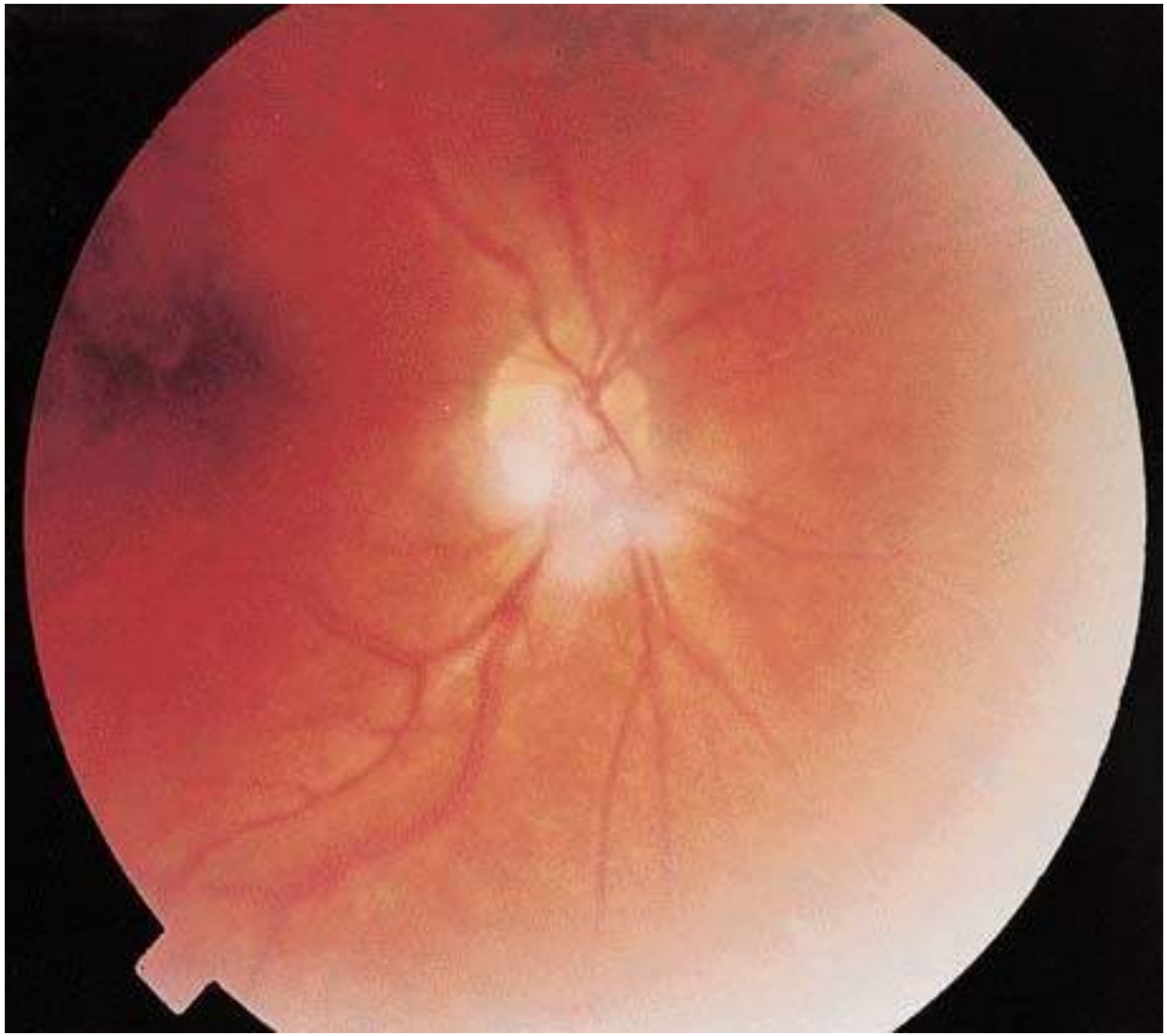
hydrolase in Lysosomes; Such deficiency in cells of fibroblasts have been demonstrated

B. Hypolipoproteinaemias

1. Abeta Lipoproteinaemia: A rare inherited disorder.
The disease is characterised by decreased plasma cholesterol↓ due to absence of β-lipoproteins (LDL). Most lipids are present in low concentration specially TG, which is virtually absent. No chylomicrons and/or pre-β lipoproteins (VLDL) are formed.

Clinical feature: Associated with above lipid changes.
There may be:

- ***Atypical relinitis pigmentosa***
- Red blood cells abnormalities like **acanthosis**
- Malabsorption of fats
- Both small intestine mucosal cells and in Liver cells, accumulation of fats occur ***(fatty infiltration)***



Metabolic defect:

Principal metabolic defect is in "synthesis of apo-B" leading to gross deficiency of apo-B resulting to deficiency of lipoproteins containing apo-B viz. chylomicrons, VLDL and LDL. Classic form of this disease is called as **Bassen-Kornzweig syndrome.**

2. Familial α-Lipoprotein Deficiency (Tangier's disease):

This disease is characterised by deficiency of α-lipoprotein (HDL↓). In homozygous patient: ***plasma HDL may be nearly completely absent.***

Inheritance: Autosomal recessive

.

Metabolic defect:

• Reduction in apo-A I and apo-A II
• Leading to accumulation of cholesteryl esters in different tissues.

Clinical features:

Clinically cardinal feature of this disease is:
• ***Hyperplastic orange yellow tonsils, and***
• ***Adenoids.***
There is no impairment of chylomicrons formation or secretion of endogenous TG by the Liver. However, on electrophoresis, there is no pre-β-lipoprotein, but a ***broad β-band is found containing the endogenous TG.***

FREDRICKSONS CLASSIFICATION OF LIPID DISORDERS

phenotype	lipoprotein	chol	TG	Clinical features
Type 1	chylomicrons	N	↑	Pancreatitis,eruptive xanthomas, lipemia retinalis
Type 11a	LDL	↑	N	Pancretitis,CAD,tendon xanthomas, arcus cornea
Type 11b	LDL+VLDL	↑	↑	Pancreatitis, CAD

phenotype	lipoprotein	chol	TG	Clinical features
Type 111	Remanant of VLDL and chylomicrons	↑	↑	Pancreatitis, CAD, PVD, palmer and tuberous xanthomas
Type 1V	VLDL	N	↑	Premature CAD
Type V	Chylomicrons+VLDL	↑	↑	Pancreatitis ,eruptive xanth, lipemia retinalis

Causes of Secondary hyperlipdemia

- Extra hepatic obstruction
- primary biliary cirrhosis
- DM
- hypothyroidism
- alcochol
- Nephrotic syndrome
- Obesity
- high dose thiazide diuretic
- exogenous sex hormone
- steroids

Familial hypercholesterolemia

- Associated with high risk of premature CAD
- Genetic defect in the LDL receptor gene resulting in either absent or defective LDL receptor activity
- This increses LDL cholesterol by 2-3 folds
- 1 in 500

Cont.

- Causes premature CAD
- Tendon xanthomas---on the backs of the hands, knuckles and the achilles tendon
- About 70 % have xanthomos by the age of 30 years
- Family screening is important
- Aggressive Rx is required

Familial combined hyperlipidemia

- Obesity and insulin resistance are common
- Accounts for about 10-15% of premature MI
- Corneal arcus are present
- Xanthelasma are present
- Tendon xanthomas DO NOT occur

Remanant particle disease

- Rare
- Risk of premature CAD and PVD
- Palmer xanthoma , tuberous and eruptive xanthomas
- Cholesterol and TG are markedly elevated

Chylomicronemia syndrome

- Characterized by massive hypertriglyceredemia
- Cholesterol level may be normal
- Usually occurs in pts with---alcohol access
  ---DM
  ---anti retroviral therapy

- Presents with----eruptive xanthomas
  ----hepatospleenomegaly
  ---lipemia retinalis
  ---↑risk of pancretitis—pts complain of intermittant abdominal pain

Risk factors

- Smoking
- Hypertension—140 / 90
- Family h/o premature CAD
- Age 45 in males---females 55
- HDL <40mg/dl
- DM

MANAGEMENT

- Main indication of RX is prevention of vascular disease
- Exclude 2ry causes—

---glucose tolerance

----TFT

---U&E

----LFT

Diet

Goal for at least 6 months

Drugs

- HMG-Co A reductase inhibitors {statins}

--pravastatin

---simvastatin

---atorvastatin

Fibrates

-gemfibrocil

--fenifibrate

Nicotinic acid

Bile acid sequesterons

--cholestyramine

---colestipol

Cholesterol absorption inhibitors

--ezetimibe

--probucol


The most pleasing five-letter word.....
"SMILE"
Keep it.


Thank
you!